

Week 1 facilitator guide: the pile game (Bulgarian solitaire)

Ten children at three tables, one adult per table, one hour. Student packets: *W1-K-v1* (K–1, 10 pages, the adult reads aloud), *W1-M-v1* (grades 2–3, 10 pages), *W1-U-v1* (grades 4–5, 13 pages). Everything the children are not meant to read is in this guide: the plan, every answer, the hints in the order to give them, and the mathematics.

1. The session at a glance

0:00–0:04	Free handling. Counters are already on the tables. Children build rows and heaps; adults say nothing about rules.
0:04–0:08	Launch by the organizer, whole group, with six large counters (Section 4): two to three minutes of talk, then the children go back to their tables.
0:08–0:50	Table work (Sections 5–7). Roughly forty minutes; each table has reserves for a fast group.
0:50–0:58	Share (Section 8). Each table reports one thing, with counters or a drawing.
0:58–1:00	Close: the organizer names the theorem and the bound (Section 8).

Tables: the parent takes K–1 (everything needed is in Section 5 and nothing has to be worked out); the second mathematician takes grades 2–3; the organizer takes grades 4–5 (the stuck $\Rightarrow$ stairs chain and Problem 8 need Section 10) and leaves that table for the launch and the close.

2. Materials and preparation

- **Counters**, flat, about 2 cm across, all one kind; at most 2 cm across for the K–1 table (its mats hold a row of four). K–1 table: 150 (the loop pages keep six pictures on the table at once, 24 to 30 counters, and the 10-counter game needs 10 more per child). Grades 2–3: 100. Grades 4–5: 100 (Problem 2(b) has three children building 21-counter staircases at once while test pictures are still out), plus **one deck of playing cards** for Problem 8 (the cards A to 7 of three suits are enough; cards slide, so use them only there). Launch: six large counters or six ordinary ones on a tray everyone can see. **One small cup per child** (10) to collect the taken counters in before the new pile is set down.
- **Packets**: three copies of the K–1 pages, four of the grades 2–3 pages, three of the grades 4–5 pages, one spare of each. Three copies of this guide.
- **Pencils** for everyone; **crayons or markers** for the K–1 table (the recording there is coloring circles). Two large blank sheets (11 $\times$ 17 in if possible) for each of the older tables, in case a map outgrows its box.
- **Preparation**: the organizer rehearses the launch (3-3 $\rightarrow$ 2-2-2 $\rightarrow$ 3-1-1-1, then stop). Each adult reads Section 3, the launch, and their own table's section, and keeps the answer tables of Section 9 at hand. The adult at K–1 also reads the K–1 answer block carefully: nothing there should be worked out during the hour.

3. The rule, the layout, the words, and the two checks

The rule. Counters sit in piles. A *move*: take exactly one counter from *every* pile (a pile of one disappears) and put all the counters you took together as one new pile. Only pile sizes matter: 3-1 and 1-3 are the same. A picture is written as its pile sizes, biggest first, hyphenated: 4-2, 3-2-1.

The layout, used at every table. Each pile is a *row* of counters; rows are stacked with the longest on top, all starting at the left. A move is then: sweep the left column into your hand, slide everything left, lay the swept counters

as a new row, and re-order the rows longest on top. Two pictures are the same exactly when the rows match one under the other, and the pictures 2-1, 3-2-1, 4-3-2-1 look like staircases. At K-1 the children may take from heaps; the adult re-lays the result as rows.

Words on the student pages. “Picture” means the arrangement of pile sizes. A picture that “stops changing” or “gives itself back” is *stuck* (a fixed point); the pages for grades 2–5 use the word stuck from their stuck-picture problem on. A game that “goes round and round” is in a *loop* (a cycle). “Stairs” or “staircase” means $k-(k-1)-\dots-1$. In this guide: n = number of counters; $T_k = 1 + 2 + \dots + k$ = the stairs numbers 1, 3, 6, 10, 15, 21, 28; a “loop” includes the case of length one when convenient, but with children keep “stuck” and “loop” separate.

Two checks, to make before every move. (i) The new pile has exactly as many counters as there were piles before the move. (ii) The number of piles after the move is (number of piles that had at least 2 counters) + 1. And the total never changes: recount when in doubt. These catch nearly every handling error.

Common errors. Forgetting that piles of 1 also give up their counter (say “one from every pile, even the tiny ones”). Putting the collected counters onto an existing pile instead of a new one. Taking from the new pile in the same move (collect into a hand or cup first, then set the new pile down). Asking “which pile do I take from?” (all of them; there are no choices, only the start). Treating 3-1 and 1-3 as different (always record largest first). Thinking “stuck” means the same counters in the same spots (it means the same sizes). When listing all pictures with 6 counters, 4-1-1 and 2-2-1-1 are the usual omissions; 11 is the check. When recording, a repeat can only be found by comparing with *all* earlier lines, not just the last one.

Pace. A move takes 40–60 seconds at K-1 with re-laying, 20–30 seconds with recording at grades 2–3, 15–20 seconds at grades 4–5; a 12-move game with 10 counters is 4–6 minutes.

4. The whole-group launch (organizer, two to three minutes of talk)

Six large counters on a tray, laid as two rows of three.

“Counters sit in piles. Here are two piles of three. A move goes like this: I take one counter from every pile” (take one from each row) “and I put the ones I took together as a new pile.” Lay them as a new row and re-sort: 2-2-2. “Again: one from every pile, three counters in my hand, they make a new pile of three.” Re-sort: 3-1-1-1. “Again?” Take one from each of the four piles, hold the four counters up, and *stop* before laying them down. “Two questions for your tables. If we keep going, will it ever stop changing? And does it matter how we start?”

Point out, while the counters are in the hand: piles of one also give up their counter and disappear; the counters in the hand always become a new pile, never join an old one; rows go longest on top so that we can tell when two pictures are the same. Do not finish the move: the next pictures would be 4-2 and then 3-2-1, and finding 3-2-1 is every table’s first job.

5. The K–1 table (two kindergartners and a first grader)

The adult reads each problem aloud, one sentence at a time, and helps with the coloring; a child who cannot yet color small circles neatly can have the adult draw dots for them while the child says what to draw. Each child has their own copy and their own counters. Core problems 1–5 take about 45 minutes; if Problem 4 runs long, the two kindergartners go to Problem 5 after start B. Problems 6–11 are reserves in order of preference. The number of moves is the number written on the first box that shows the stairs (box 0 is the start; the last colored box only repeats it). In Problem 3, where the boxes start at 1, it is the number of the first stairs box.

Order and timing

Problem	Minutes	What it does
1: 3 counters from 3	4	The move, and the first stuck picture 2-1.
2: 3 counters from 1-1-1	3	A different start, the same stuck picture; piles of 1 vanish.
3: 6 counters from 6	6	Three moves to 3-2-1; the outline is a staircase.
4: 6 counters, own starts	20	Every start ends at 3-2-1. Move counts differ. Two starts for the kindergartners, three for the first grader.
5: 4 counters from 4	8	Round and round: three pictures repeat forever, no stairs.
6: build the stairs	5	The stairs numbers 1, 3, 6, 10, 15; 4-3-2-1 is stuck.
7: 10 counters from 10	8	Six moves to 4-3-2-1; a small staircase grows under the long row. First grader first.
8: 5 counters from 5	6	Round and round again, with three pictures.
9–10: 2 and 1 counters; the chart	5	Which numbers stop: 1, 3, 6, 10. Which go round: 2, 4, 5, 7.
11: 7 counters from 7	8	A loop of four pictures; the one extra counter walks down the stairs.

Answers

- **Problem 1.** $3 \rightarrow 2-1 \rightarrow 2-1 \rightarrow 2-1$. Boxes 1, 2 and 3 all show 2-1 (two colored circles in the top row, one below). Any two boxes look the same; the game stopped changing after the first move. Say “stuck” here and use it from now on.
- **Problem 2.** $1-1-1 \rightarrow 3 \rightarrow 2-1 \rightarrow 2-1$. Box 1 shows one row of three, boxes 2 and 3 show 2-1. It stopped changing at box 2, and the stuck picture is the same as in Problem 1 even though the start was different.
- **Problem 3.** $6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1 \rightarrow 3-2-1$. Boxes 1–4: 5-1, 4-2, 3-2-1, 3-2-1. The line around the last box is a staircase: three steps. Name it: “the stairs”.
- **Problem 4.** Every one of the eleven pictures with 6 counters ends at 3-2-1. Moves to the stairs: 6: 3; 5-1: 2; 4-2: 1; 4-1-1: 5; 3-3: 4; 3-2-1: 0; 3-1-1-1: 2; 2-2-2: 3; 2-2-1-1: 6; 2-1-1-1-1: 3; 1-1-1-1-1-1: 4. The games: $3-3 \rightarrow 2-2-2 \rightarrow 3-1-1-1 \rightarrow 4-2 \rightarrow 3-2-1$; 2-2-2 joins that game at its second picture; $1-1-1-1-1-1 \rightarrow 6 \rightarrow 5-1 \rightarrow 4-2 \rightarrow 3-2-1$; $4-1-1 \rightarrow 3-3 \rightarrow \dots$; $2-2-1-1 \rightarrow 4-1-1 \rightarrow 3-3 \rightarrow \dots$; $2-1-1-1-1 \rightarrow 5-1 \rightarrow \dots$. The box after the first 3-2-1 shows 3-2-1 again. Suggest the printed starts so that the three children take different ones; a child who wants “the longest” gets 2-2-1-1. In the big box on page 4 both drawings are 3-2-1: same.
- **Problem 5.** $4 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1$. Mats 1–6: 3-1, 2-2, 2-1-1, 3-1, 2-2, 2-1-1. Pairs: 1 and 4, 2 and 5, 3 and 6. It never gets stuck: it goes round 3-1, 2-2, 2-1-1, 3-1, ... forever. Ask: “What would mat 7 show? And mat 100?” (Mat 7 is 3-1 again; the pattern repeats in threes.) This is the K–1 share item: 6 makes stairs, 4 goes round.

- **Problem 6.** The staircases hold 1, 3, 6 and 10 counters; the next one, 5-4-3-2-1, holds 15. A move on 4-3-2-1: the rows become 3, 2, 1, nothing, and the four counters in the hand make a new row of 4: 4-3-2-1 again, so it is stuck. (Optional: the same on 5-4-3-2-1, which is also stuck.)
- **Problem 7.** $10 \rightarrow 9-1 \rightarrow 8-2 \rightarrow 7-2-1 \rightarrow 6-3-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1 \rightarrow 4-3-2-1$. Boxes 1–7, with box 7 equal to box 6 (box 8 stays empty). The rows under the top row are 1; 2; 2-1; 3-1; 3-2; 3-2-1; 3-2-1: a small staircase grows down there while the long row shrinks, and at the end the long row is exactly one step longer than the staircase under it. From 5-5 it also takes six moves: $5-5 \rightarrow 4-4-2 \rightarrow 3-3-3-1 \rightarrow 4-2-2-2 \rightarrow 4-3-1-1-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$.
- **Problem 8.** $5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2 \rightarrow 2-2-1$. Mats 1–6: 4-1, 3-2, 2-2-1, 3-1-1, 3-2, 2-2-1. Pairs: 2 and 5, 3 and 6 (mat 1 has no partner: 4-1 is passed once and never again). Three pictures go round: 3-2, 2-2-1, 3-1-1.
- **Problems 9 and 10.** $2 \rightarrow 1-1 \rightarrow 2 \rightarrow 1-1$: boxes 1–3 show 1-1, 2, 1-1; two pictures go round. One counter: the row of one is taken and the one counter in the hand makes a new row of one, so box 4 shows a single circle; 1 is stuck. Chart: 1 stops, 2 round, 3 stops, 4 round, 5 round, 6 stops, 7 round (Problem 11), 10 stops (Problem 7). The numbers that stop are the staircase numbers 1, 3, 6, 10. The stairs icon means “stops”; for 1 counter the stairs is a single counter, which a child may object to: agree, and circle it anyway.
- **Problem 11.** $7 \rightarrow 6-1 \rightarrow 5-2 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2$. Boxes 1–9; box 3 equals box 7, box 4 equals box 8, box 5 equals box 9. Four pictures go round. In boxes 3–6 the stairs 3-2-1 is there with one extra circle: at the end of the top row (4-2-1), at the end of the second row (3-3-1), at the end of the third row (3-2-2), and as a fourth row of its own (3-2-1-1). The extra counter walks down one row per move and, after stepping off the bottom, reappears at the top.

Hints, in order

1. Before each move: “How many rows are there? So how many counters will be in your hand? So how long is the new row?” Then check.
2. “One from every row, even a row with only one.” Watch the vanishing rows of one; some children keep the counter and the row.
3. “Put the ones in your hand down as their own new row, then put the longest row on top.”
4. Same or different: “Put this picture next to the one before, row under row. Same?” For the loop pages: “Have we seen this picture already? Look at all the mats.”
5. Recount: “Still six? So nothing got lost.”

If the table stalls, or finishes early

A child who loses track mid-game rebuilds the last colored box with counters and continues from there. If coloring is slow, the adult colors and the child dictates (“two in the top row, one under”). If a kindergartner tires of 6 counters in Problem 4, let that child go to Problem 5 (the mats) while the first grader finishes a third start. Finishing early: Problems 6–11 in the printed order (Problem 7 is for the first grader, Problem 8 for the kindergartners while they wait). After that: 8 counters from 8 on blank paper as mats, $8 \rightarrow 7-1 \rightarrow 6-2 \rightarrow 5-2-1 \rightarrow 4-3-1 \rightarrow 3-3-2 \rightarrow 3-2-2-1 \rightarrow 4-2-1-1 \rightarrow 4-3-1$, a loop of four after four moves; or build the biggest staircase the tub allows and count it (21, 28).

Complete maps for the K–1 adult

For 3 counters: $1-1-1 \rightarrow 3 \rightarrow 2-1$ (stuck). For 4 counters: $1-1-1-1 \rightarrow 4 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1$, so every start is in the loop of three within two moves. For 5 counters: $1-1-1-1-1 \rightarrow 5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2$, and

2-1-1-1 → 4-1; the loop is 3-2, 2-2-1, 3-1-1. For 6 counters, all eleven games are listed under Problem 4 above, and the map is drawn in Section 9. For 7 counters: the single-pile game under Problem 11; the loop is 4-2-1, 3-3-1, 3-2-2, 3-2-1-1, and every one of the 15 pictures reaches it within four moves.

6. The grades 2–3 table (four third graders)

Children record in numbers, biggest pile first. The adult keeps the answer tables of Section 9 open, especially the 42-row table for 10 counters. At 20–30 seconds per recorded move, the core is Problems 1–4 and the 4-counter half of Problem 5 (about 40 minutes). The reserves are taken *in the order of the table below, not in page order*: the 5-counter half of Problem 5, then Problems 6, 8, 9–10, 7, 11, 12. The share item (the Y for 6 and the loop for 4) comes from the core.

Order and timing

Problem	Minutes	What it does
1: 6 counters, own starts	10	Everyone reaches 3-2-1; move counts differ (1 to 6). Three games, about 15 lines.
2–3: list all pictures, and their next pictures	10	Eleven pictures; the table of all cases; eleven moves.
4: the map for 6	10	A Y-shaped river: every arrow path ends at 3-2-1. Longest route 2-2-1-1, 6 moves. Three sources.
5, 4-counter half	8	A number with no stuck picture: a loop of three, shown by exhaustion.
Reserves, in this order:		
5, 5-counter half	8	The same with seven pictures; the loop again has three.
6: which pictures are stuck	12	Stairs, and only stairs; the stairs numbers 1, 3, 6, 10, 15, 21.
8: 8 counters, two loops	10	Not every start ends in the same place.
9–10: 7 and 9 counters	12	Loop pictures are stairs plus wandering extras.
7: 10 counters	15	All 42 starts reach 4-3-2-1; 10 takes 6, the longest 12.
11: predict before you move	6	The two checks, as a game.
12: pictures nothing leads to	6	Undoing a move; Garden of Eden pictures.

Answers

- **Problem 1.** Every start reaches 3-2-1, which gives itself back. Moves: 6: 3; 5-1: 2; 4-2: 1; 4-1-1: 5; 3-3: 4; 3-2-1: 0; 3-1-1-1: 2; 2-2-2: 3; 2-2-1-1: 6; 2-1-1-1-1: 3; 1-1-1-1-1-1: 4. The full games are under K–1 Problem 4. Steer the four children to four different starts; someone should have 2-2-1-1 or 4-1-1.
- **Problems 2–3.** The eleven pictures and their next pictures: 6 → 5-1; 5-1 → 4-2; 4-2 → 3-2-1; 4-1-1 → 3-3; 3-3 → 2-2-2; 3-2-1 → 3-2-1; 3-1-1-1 → 4-2; 2-2-2 → 3-1-1-1; 2-2-1-1 → 4-1-1; 2-1-1-1-1 → 5-1; 1-1-1-1-1-1 → 6. Completeness: “Biggest pile 4: what can the other two counters be? 2 or 1-1. Biggest pile 3: the other three are 3, 2-1 or 1-1-1.”
- **Problem 4.** The map is the Y in Section 9. (a) 2-2-1-1, 6 moves. (b) 2-2-1-1, 2-1-1-1-1, 1-1-1-1-1-1. (c) 5-1 (from 6 and 2-1-1-1-1), 4-2 (from 5-1 and 3-1-1-1), 3-2-1 (from 4-2 and from itself). The argument to draw out: “You listed every picture and every arrow. Could a game with 6 counters avoid 3-2-1? Point to where it could escape.” There is nowhere: every path flows into 3-2-1.

- **Problem 5.** 4 counters, five pictures: $4 \rightarrow 3-1$; $3-1 \rightarrow 2-2$; $2-2 \rightarrow 2-1-1$; $2-1-1 \rightarrow 3-1$; $1-1-1-1 \rightarrow 4$. No stuck picture; the loop is $3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1$. 5 counters, seven pictures: $5 \rightarrow 4-1$; $4-1 \rightarrow 3-2$; $3-2 \rightarrow 2-2-1$; $2-2-1 \rightarrow 3-1-1$; $3-1-1 \rightarrow 3-2$; $2-1-1-1 \rightarrow 4-1$; $1-1-1-1-1 \rightarrow 5$. Loop $3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2$. The printed question: no picture with 4 counters gives itself back, and none with 5; the list shows it because every picture is there with its next picture, and no row has the same picture in both columns. Why a loop is forever: “If a picture comes back, what happens next? The rule does the same thing to the same picture.” Expect “2-2 is stuck”: test it ($\rightarrow 2-1-1$).
- **Problem 6.** 2-1: stuck. 3-2-1: stuck. 4-3-2-1: stuck. $2-2 \rightarrow 2-1-1$: no. $4-2 \rightarrow 3-2-1$: no. $3-3 \rightarrow 2-2-2$: no. $5-4-3-2-1$: stuck. $6-3-1 \rightarrow 5-3-2$: no. (b) Guesses to expect and test: 4-2, 2-2, 3-3, 4-4-2; none is stuck. (c) Counts 3, 6, 10, 15 (and 1 if someone tests a single counter). (d) The next stuck picture 6-5-4-3-2-1 has 21. (e) Why 3-2-1 is stuck: three becomes two, two becomes one, one disappears, and the three counters taken make a three. For 4-3-2-1: each row shrinks by one and becomes the row below it, the bottom row disappears, and the four counters taken make the new top row. (Only staircases are stuck: the reason is in Section 10, and the grades 4–5 table produces it.)
- **Problem 7.** A: $10 \rightarrow 9-1 \rightarrow 8-2 \rightarrow 7-2-1 \rightarrow 6-3-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$, 6 moves. B: $5-5 \rightarrow 4-4-2 \rightarrow 3-3-3-1 \rightarrow 4-2-2-2 \rightarrow 4-3-1-1-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$, 6 moves. C: $2-2-2-2-2 \rightarrow 5-1-1-1-1-1 \rightarrow 6-4 \rightarrow 5-3-2 \rightarrow 4-3-2-1$, 4 moves. D: see the 42-row table in Section 9; every start reaches 4-3-2-1. Starts with more than 8 moves: 9 moves from 8-1-1, 4-3-3, 4-1-1-1-1-1-1; 10 from 5-4-1, 4-4-1-1, 2-2-1-1-1-1-1-1; 11 from 6-2-1-1, 5-2-2-1, 5-2-1-1-1; 12 from 3-3-2-1-1, 3-2-2-2-1, 3-2-2-1-1-1. The single pile is not the slowest start.
- **Problem 8.** A: $8 \rightarrow 7-1 \rightarrow 6-2 \rightarrow 5-2-1 \rightarrow 4-3-1 \rightarrow 3-3-2 \rightarrow 3-2-2-1 \rightarrow 4-2-1-1 \rightarrow 4-3-1$: loop A is $4-3-1 \rightarrow 3-3-2 \rightarrow 3-2-2-1 \rightarrow 4-2-1-1 \rightarrow$ back, four pictures. B: $2-2-2-2 \rightarrow 4-1-1-1-1 \rightarrow 5-3 \rightarrow 4-2-2 \rightarrow 3-3-1-1 \rightarrow 4-2-2$: loop B is $4-2-2 \rightarrow 3-3-1-1 \rightarrow$ back, two pictures. Not the same loop. $5-3 \rightarrow 4-2-2$: B. $4-4 \rightarrow 3-3-2$: A. $6-1-1 \rightarrow 5-3$: B. Only seven of the 22 starts reach B: 2-2-2-2, 2-2-1-1-1-1, 4-1-1-1-1, 6-1-1, 5-3, 4-2-2, 3-3-1-1; the other fifteen reach A. Share item: with 8 counters it *does* matter how you start.
- **Problem 9.** $7 \rightarrow 6-1 \rightarrow 5-2 \rightarrow 4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow 4-2-1$. Loop: 4-2-1, 3-3-1, 3-2-2, 3-2-1-1. The ringed dot is in row 1, row 2, row 3, and then row 4, a row of its own under the stairs (blanks: 1, 2, 3, 4): it walks down one row per move and jumps back to the top.
- **Problem 10.** $9 \rightarrow 8-1 \rightarrow 7-2 \rightarrow 6-2-1 \rightarrow 5-3-1 \rightarrow 4-3-2 \rightarrow 3-3-2-1 \rightarrow 4-2-2-1 \rightarrow 4-3-1-1 \rightarrow 4-3-2$. Loop: 4-3-2, 3-3-2-1, 4-2-2-1, 4-3-1-1. Extras beside rows 1, 2, 3; then 2, 3, 4; then 1, 3, 4; then 1, 2, 4. All three step down one row at once, and the one that steps off the bottom reappears at the top.
- **Problem 11.** New pile = number of piles; piles after = (piles with at least 2) + 1. 5-3-2: 3, 4, 4-3-2-1. 4-4-1-1: 4, 3, 4-3-3. 6-1-1-1: 4, 2, 5-4. 2-2-2-2-2: 5, 6, 5-1-1-1-1-1. 3-3-2-1-1: 5, 4, 5-2-2-1. 7-1: 2, 2, 6-2. 3-2-2-1: 4, 4, 4-2-1-1. 5-1-1-1-1-1: 6, 2, 6-4.
- **Problem 12.** (a) If six piles of 1 had just been made, the new pile has 1 counter, and the new pile is always as big as the number of piles before the move (Problem 11), so there was exactly one pile before: 6. But $6 \rightarrow 5-1$, not $1-1-1-1-1-1$. So nothing leads to six singles. (b) For 2-2-1-1: there are four piles, so the new pile must have handed one counter to each of the other three and needs at least 3 counters; a pile of 2 runs out after two piles, a pile of 1 after one. Nothing leads to it. (c) 2-1-1-1-1: five piles, the new pile would need 4; no pile has 4. Nothing leads to it either. These three are exactly the pictures with no arrow into them on the map.

Hints, in order

1. “Before you move, tell me the size of the new pile. Now check.” Then “how many piles will there be?”
2. Recording: one line per move, biggest first; after each line, recount the counters.
3. Repeats: “Compare with every earlier line, not just the last one.”
4. Listing all pictures: “Biggest pile 6 first. Then biggest pile 5: what goes with it? Then 4 . . .” Check the count, 11.

5. The map: “Put 3-2-1 at the bottom. Which pictures point straight at it? Draw those next.”
6. Stuck: “Say out loud what happens to each pile.”

If the table stalls, or finishes early

If the listing in Problem 2 stalls at 9 or 10, ask for the missing biggest-pile group (usually 4 or 2). If the map is a tangle, redraw it on a large sheet with 3-2-1 at the bottom and the two branches left and right (Section 9). If the core runs slow, the 5-counter half of Problem 5 is the first thing to drop; the share item needs only the map for 6 and the 4-counter loop. When Problem 7 is reached and drags, stop after games A and B and give the fact: all 42 starts reach 4-3-2-1, the longest in 12 moves. Finishing early: the reserves in the order of the table (5-counter half of 5, then 6, 8, 9–10, 7, 11, 12); then “find a start with 10 counters that takes exactly 12 moves” (three exist), or “is there a stuck picture with 12 counters?” (no; let them search, then say that the grades 4–5 table has the reason), or the 15-counter single-pile game on paper (10 moves, Section 9).

7. The grades 4–5 table (two fourth graders and a fifth grader)

At 15–20 seconds per move, the core is Problem 1, Problem 2 and Problem 3 (a)–(d) (about 40 minutes). If Problem 3 runs long, stop after (d) and share loops A and B from the table without the map. The reserves are taken *in the order of the table below, not in page order*: 3 (e)(f), then Problems 4, 6, 5, 7, 8, 9. Problem 8 needs the cards and an adult who has read Section 10, and is handed out only after hint 7 (“Is there a reason that does not need the list?”) has been asked.

Order and timing

Problem	Minutes	What it does
1: the full map for 6	13	Eleven pictures, all reaching 3-2-1; longest 6 from 2-2-1-1; the map and (d)–(g).
2: stuck pictures	12	Stuck means stairs, with the argument; stairs numbers; the 12-counter search.
3 (a)–(d): the 22 pictures for 8, two loops	15	22 listings and moves; two loops, basins of 15 and 7.
Reserves, in this order:		
3 (e)(f): the map for 8, the rule hunt	12	The two-basin map; no simple rule.
4: 10 counters and 15	20	Exact counts: 6, 7, 12; patterns T_{k-1} and $k(k-1)$; the 15-pile game.
6: undoing moves	10	Garden of Eden pictures and the rule for them.
5: loops for other numbers	25	Loop lengths 3, 3, 4, 4 and 2, 4, 5, 5; stairs plus wanderers.
7: how many pictures	8	1, 2, 3, 5, 7, 11, 15, 22 and the completeness argument.
8: cards on diagonals	15–18	The invariant behind the theorem, seen twice.
9: 12 counters	15	Two loops of five; every loop for 12 has length 5.

Answers

- **Problem 1.** The eleven pictures with their next picture and moves to 3-2-1: $6 \rightarrow 5-1, 3$; $5-1 \rightarrow 4-2, 2$; $4-2 \rightarrow 3-2-1, 1$; $4-1-1 \rightarrow 3-3, 5$; $3-3 \rightarrow 2-2-2, 4$; $3-2-1 \rightarrow 3-2-1, 0$; $3-1-1-1 \rightarrow 4-2, 2$; $2-2-2 \rightarrow 3-1-1-1, 3$; $2-2-1-1$

→ 4-1-1, 6; 2-1-1-1-1 → 5-1, 3; 1-1-1-1-1-1 → 6, 4. The map is the Y in Section 9. (e) 2-2-1-1, 2-1-1-1-1, 1-1-1-1-1-1. (f) 2-2-1-1 → 4-1-1 → 3-3 → 2-2-2 → 3-1-1-1 → 4-2 → 3-2-1, 6 moves. (g) No: every picture is on the map, every picture has exactly one arrow out, and following arrows from anywhere ends at 3-2-1; a game that ran forever would have to leave the map or find a second loop, and there is none.

- **Problem 2.** (a) 3-2-1 yes; 2-2 → 2-1-1 no; 4-2 → 3-2-1 no; 3-3 → 2-2-2 no; 4-3-2-1 yes; 6-3-1 → 5-3-2 no; 5-4-3-2-1 yes; 4-4-2 → 3-3-3-1 no. (b) 5-4-3-2-1 (15) and 6-5-4-3-2-1 (21). (c) Nothing works: 12 is not a stairs number, and the only stuck pictures are stairs. (d) After the move every old pile is smaller than it was, so no old pile has 5 any more; the pile of 5 in the new picture must be the new pile. The new pile's size is the number of piles, so there are 5 piles. The old 5 shrank to 4, and since the picture did not change there is a 4; that 4 shrinks to a 3, so there is a 3; likewise a 2 and a 1. Five piles of sizes 5, 4, 3, 2, 1: 5-4-3-2-1, which is stuck. (e) Each row of 3-2-1 shrinks by one and becomes the row below it, the bottom row disappears, and the three counters taken make a new top row of three; with 10 rows, the same, with ten counters taken. (f) 1, 2-1, 3-2-1, 4-3-2-1, 5-4-3-2-1, 6-5-4-3-2-1: 1, 3, 6, 10, 15, 21 counters; the next stairs has 28. The full argument (stuck ⇒ stairs), to lead in this order: "In a stuck picture, call the biggest pile B . After the move, is any old pile still as big as B ? So which pile is the size- B pile of the new picture? How big is the new pile always? So B equals what? The old B -pile shrank to what? Is that size in the picture? Keep going: $B, B-1, \dots$, down to what? How many piles is that? How many piles are there? So what are all the piles?"
- **Problem 3.** The 22 pictures, next picture, and loop: biggest 5 to 8: 8 → 7-1 A; 7-1 → 6-2 A; 6-2 → 5-2-1 A; 6-1-1 → 5-3 B; 5-3 → 4-2-2 B; 5-2-1 → 4-3-1 A; 5-1-1-1 → 4-4 A. Biggest 4 or 3: 4-4 → 3-3-2 A; 4-3-1 → 3-3-2 A; 4-2-2 → 3-3-1-1 B; 4-2-1-1 → 4-3-1 A; 4-1-1-1-1 → 5-3 B; 3-3-2 → 3-2-2-1 A; 3-3-1-1 → 4-2-2 B; 3-2-2-1 → 4-2-1-1 A; 3-2-1-1-1 → 5-2-1 A; 3-1-1-1-1-1 → 6-2 A. Biggest 2 or 1: 2-2-2-2 → 4-1-1-1-1 B; 2-2-2-1-1 → 5-1-1-1 A; 2-2-1-1-1-1 → 6-1-1 B; 2-1-1-1-1-1-1 → 7-1 A; 1-1-1-1-1-1-1-1 → 8 A. Total 22 (7, 10, 5). Loop A: 4-3-1 → 3-3-2 → 3-2-2-1 → 4-2-1-1 → back (length 4), reached from 8 after 4 moves. Loop B: 4-2-2 → 3-3-1-1 → back (length 2), reached from 2-2-2-2 after 3 moves. Basins: A has 15 pictures, B has 7: 2-2-2-2, 2-2-1-1-1-1, 4-1-1-1-1, 6-1-1, 5-3, 4-2-2, 3-3-1-1. The map is drawn in Section 9. (f) No simple rule is known to us; every natural guess breaks: "all piles even" (2-2-2-2 is B but 4-4 and 6-2 are A); "two piles" (5-3 is B, 4-4 is A); "biggest pile even" (4-2-2 is B, 4-3-1 is A); "an odd number of piles" (6-1-1 is B, 4-3-1 is A). The parity of the number of piles, of the biggest pile, all-even, and the potential of Section 10 have all been checked and none separates the basins. The honest statement: the table they built *is* the rule; a shorter one is an open question for them.
- **Problem 4.** (a) 10 → 9-1 → 8-2 → 7-2-1 → 6-3-1 → 5-3-2 → 4-3-2-1: 6 moves; the stairs grows from the small end. (b) Ten singles → 10, then as in (a): 7 moves. (c) 3-3-2-1-1 → 5-2-2-1 → 4-4-1-1 → 4-3-3 → 3-3-2-2 → 4-2-2-1-1 → 5-3-1-1 → 4-4-2 → 3-3-3-1 → 4-2-2-2 → 4-3-1-1-1 → 5-3-2 → 4-3-2-1: 12 moves, the longest possible. (d) The 42-row table in Section 9. (e) Moves from one pile: 1: 0; 3: 1; 6: 3; 10: 6; 15: 10 (the stairs numbers again, one step behind). Longest game: 1: 0; 3: 2 (from 1-1-1); 6: 6 (from 2-2-1-1); 10: 12; 15: 20 (from 4-4-3-2-1-1, among 16 starts). A child may see 0, 2, 6, 12, 20 = $1 \cdot 0, 2 \cdot 1, 3 \cdot 2, 4 \cdot 3, 5 \cdot 4$. (f) 15 → 14-1 → 13-2 → 12-2-1 → 11-3-1 → 10-3-2 → 9-3-2-1 → 8-4-2-1 → 7-4-3-1 → 6-4-3-2 → 5-4-3-2-1: 10 moves. (g) Eleven moves from 6-2-1-1, 5-2-2-1, 5-2-1-1-1; twelve from 3-3-2-1-1, 3-2-2-2-1, 3-2-2-1-1-1; ten from 5-4-1, 4-4-1-1, 2-2-1-1-1-1-1-1.
- **Problem 5.** (a) 4 from 4: loop 3-1, 2-2, 2-1-1, length 3. 5 from 5: 3-2, 2-2-1, 3-1-1, length 3. 7 from 7: 4-2-1, 3-3-1, 3-2-2, 3-2-1-1, length 4. 8 from 8: 4-3-1, 3-3-2, 3-2-2-1, 4-2-1-1, length 4. 8 from 2-2-2-2: 4-2-2, 3-3-1-1, length 2. 9 from 9 (8-1, 7-2, 6-2-1, 5-3-1, then the loop): 4-3-2, 3-3-2-1, 4-2-2-1, 4-3-1-1, length 4. 11 from 11 (10-1, 9-2, 8-2-1, 7-3-1, 6-3-2, then the loop): 5-3-2-1, 4-4-2-1, 4-3-3-1, 4-3-2-2, 4-3-2-1-1, length 5. 12 from 12 (11-1, 10-2, 9-2-1, 8-3-1, 7-3-2, 6-3-2-1, then the loop): 5-4-2-1, 4-4-3-1, 4-3-3-2, 4-3-2-2-1, 5-3-2-1-1, length 5. (b) For 7 the extra dot sits at the end of row 1, 2, 3, then forms row 4. (c) For 11 the extra sits at the end of row 1, 2, 3, 4, then forms row 5 (4-3-2-1-1). (d) For 9 the extras are beside rows 1, 2, 3; then 2, 3, 4; then 1, 3, 4; then 1, 2, 4: all step down together, and the one that leaves the bottom reappears at the top. The pattern to name: with $T_k + r$ counters ($0 \leq r \leq k$), every loop picture is the stairs with k rows plus r extra counters, at most one per row (a phantom empty row $k+1$ counts as a row), and each move shifts all extras down one row, cyclically. So the loop length divides $k+1$: 3 for 4 and 5 counters ($k=2$), 4 for 7, 8, 9 ($k=3$; the pattern "rows 1 and 3" at $n=8$ has period 2), 5 for 11 to 14.

- **Problem 6.** (a) 5-2 (the 2 was new: it gave one to the 4 and one to the 1) and 3-2-1-1 (the 4 was new: it gave one each to the 2 and the 1 and had two left over as vanished singles); the 1 cannot be new, it would have to give away two. (b) 6-4, 6-3-1, 4-3-1-1-1. (c) 4-2 and 3-2-1 itself. (d) 3-2-1-1-1, 3-1-1-1-1-1, 2-2-2-2, 2-2-2-1-1, 2-2-1-1-1-1, 2-1-1-1-1-1-1, 1-1-1-1-1-1-1-1: seven of them. (e) 3-3-3: yes, from 4-4-1. 2-2-2-2-1: no (five piles; the new pile would need 4). 5-1-1-1-1-1-1: no (seven piles; needs 6). 4-4-4-4-4: yes, from 5-5-5-5. 3-1-1-1-1-1-1: no (seven piles; needs 6). 9-1-1-1-1-1-1-1-1-1: yes, from nine piles of 2. (f) Some picture leads to a given picture exactly when its biggest pile is at least (number of piles) -1 ; the number of pictures leading to it is the number of *distinct* pile sizes that are at least (number of piles) -1 .
- **Problem 7.** 1, 2, 3, 5, 7, 11, 15, 22. The box holds the lists for 4 (five pictures: 4, 3-1, 2-2, 2-1-1, 1-1-1-1), 5 (seven: 5, 4-1, 3-2, 3-1-1, 2-2-1, 2-1-1-1, 1-1-1-1-1) and 7 (below); 6 and 8 come from Problems 1 and 3, and 1, 2, 3 are done in the head. Completeness: list by biggest pile; within one biggest pile, the rest is a picture of a smaller number whose biggest pile is no larger, so the lists for smaller numbers do the work. The 15 pictures with 7 counters: 7; 6-1; 5-2, 5-1-1; 4-3, 4-2-1, 4-1-1-1; 3-3-1, 3-2-2, 3-2-1-1, 3-1-1-1-1; 2-2-2-1, 2-2-1-1-1, 2-1-1-1-1-1; 1-1-1-1-1-1-1.
- **Problem 8.** Start 4-4-2: rows A 2 3 4 / 2 3 4 5 / 3 4. Grid 1 (3-3-3-1): A 2 3 / 2 3 4 / 3 4 5 / 4. Grid 2 (4-2-2-2): A 2 3 4 / 2 3 / 3 4 / 4 5. Grid 3 (4-3-1-1-1): A 2 3 4 / 2 3 4 / 3 / 4 / 5. Grid 4 (5-3-2): A 2 3 4 5 / 2 3 4 / 3 4. In all of these every card satisfies rank = row + column -1 : each card just moves along its diagonal, and no re-sorting was ever needed because the new top row was always the longest. Grid 5 (rows 3, 4, 2, 1, out of order): A 2 3 / 2 3 4 5 / 3 4 / 4. Sorting swaps the top two rows: grid 6 is 4-3-2-1 with rows 2 3 4 5 / A 2 3 / 3 4 / 4. (c) The seven cards of the two swapped rows break the rule: the four cards of the new top row are each one too big, the three cards of the new second row each one too small. (d) Grid 5 total 31, grid 6 total 30, difference 1: the swap moved four cards up a diagonal and three cards down. Say clearly that the pictures are the same as in the ordinary game; only where the new row is laid differs. (e) Start 7-1-1-1: rows A 2 3 4 5 6 7 / 2 / 3 / 4, every card obeying the rule, total 37. The special move takes the left column A, 2, 3, 4 and lays it as the new top row; the old top row slides to 2 3 4 5 6 7 and rows 2–4 vanish. Grid 7: A 2 3 4 / 2 3 4 5 6 7, lengths 4 and 6, out of order at once (the new row of 4 is shorter than $7 - 1$), every card still on its diagonal, total still 37. Sorting swaps the two rows: grid 8 is 2 3 4 5 6 7 / A 2 3 4, the picture 6-4. All ten cards break the rule: the six of the new top row are each one too big, the four of the second row each one too small. Totals 37 and 35, difference $2 = 6 - 4$. Blanks at the bottom: grid 6 came from swapping rows of 4 and 3, drop 1; grid 8 from rows of 6 and 4, drop 2. The contrast to state: the special move never changes the total, and each swap of a short row above a longer one drops it by (long row minus short row); so the total never rises, and it stays put exactly when no sorting is needed. This is the proof of Section 10, seen twice.
- **Problem 9.** From 12: 11-1, 10-2, 9-2-1, 8-3-1, 7-3-2, 6-3-2-1, then loop A: 5-4-2-1 $\rightarrow$ 4-4-3-1 $\rightarrow$ 4-3-3-2 $\rightarrow$ 4-3-2-2-1 $\rightarrow$ 5-3-2-1-1 $\rightarrow$ back. From 2-2-2-2-2-2: 6-1-1-1-1-1-1, 7-5, 6-4-2, then loop B: 5-3-3-1 $\rightarrow$ 4-4-2-2 $\rightarrow$ 4-3-3-1-1 $\rightarrow$ 5-3-2-2 $\rightarrow$ 4-4-2-1-1 $\rightarrow$ back. Different loops, both of length 5. (b) With the stairs 4-3-2-1 and a phantom row 5, the extras of loop A sit in rows 1,2; 2,3; 3,4; 4,5; 5,1; those of loop B in rows 1,3; 2,4; 3,5; 4,1; 5,2. (c) Every one of the 77 pictures with 12 counters reaches loop A or loop B (45 reach A, 32 reach B), so every loop has length 5; there is no other. Let the search run for two or three starts (each is about ten moves), then give the fact and hand them the stairs-plus-wanderers reason: every loop picture is 4-3-2-1 plus two extras, the extras step down one row per move through five rows (the fifth is the empty one under the stairs), and two extras come back to their places after five steps whatever their spacing. (After 8, the next number of counters with loops of two different lengths is 17: lengths 6, 6 and 3.)

Hints, in order

1. The two checks before every move (new pile size; number of piles after).
2. Listing: by biggest pile, then by the next pile; “how do you know there is not a twelfth?”
3. The map: bottom-up from the stuck picture, or loops first and then feeders.

4. Stuck $\Rightarrow$ stairs: the question chain under Problem 2.
5. Wanderers: “Draw the stairs and mark the extra. Move. Where is the extra now? Again. What happens when it reaches the bottom?”
6. Undoing: “Pretend this pile is the new one. Give one back to every other pile. Enough counters? Try the next pile.”
7. Beyond exhaustion: “You proved 6 by listing all cases, and 8 the same way. For 10 there are 42 pictures, for 15 there are 176, for 21 there are 792. Is there a reason that does not need the list?” Lead to the wanderers and, if there is time, to Problem 8; otherwise leave it open and name the theorem.

If the table stalls, or finishes early

If the 8-map becomes a tangle, move to a large sheet: draw loop A as a square and loop B as a pair, then attach feeders from the table. If Problem 3(f) frustrates, say that no short rule is known to us and that the table is the answer; move on. If the argument in Problem 2(d) does not come, give it and ask them to repeat it for biggest pile 6. If the core runs slow, Problem 3 stops after (d) and the share uses the loops from the table. Finishing early: the reserves in the order of the table (3 (e)(f), then 4, 6, 5, 7, 8, 9); then the questions at the end of Section 10 (which starts with 10 counters take exactly 12 moves and why they all pass through 4-3-3; the distribution of move counts for 10 counters; whether 21 counters can take 30 moves).

8. Share and close (ten minutes)

Each table speaks for two minutes, with counters or a drawing held up.

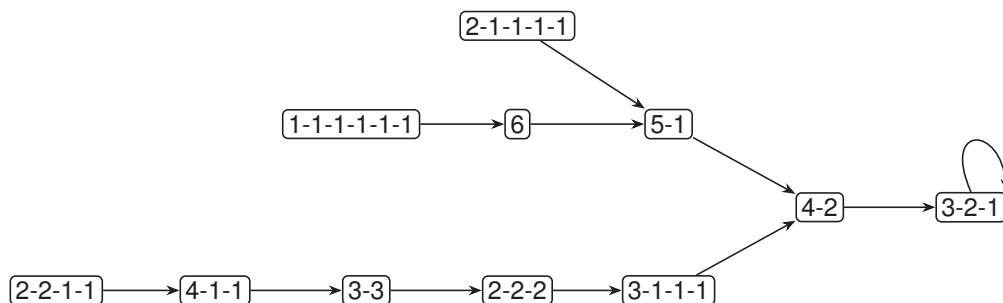
- **K–1**: “6 makes stairs, 4 goes round.” A child lays 3-2-1 and shows that a move gives it back; another shows the three mats of the 4-counter loop.
- **Grades 2–3**: the map for 6 (the Y, all arrows into 3-2-1) and the loop for 4. One sentence: “We checked every picture, so no start with 6 counters can avoid the stairs.”
- **Grades 4–5**: which pictures are stuck (stairs only) and why, in the children’s words; and that with 8 counters two different loops happen, so the start does matter.

The organizer closes: “What you found is a theorem. Jørgen Brandt proved in 1982 that when the number of counters is a stairs number, every start ends at the stairs, and when it is not, every start ends in a loop, and the loop pictures are always a stairs with a few extra counters walking down it. Kiyoshi Igusa and Gwihen Étienne then showed that with $1 + 2 + \dots + k$ counters the stairs always arrives within $k \cdot (k - 1)$ moves: 0, 2, 6, 12, 20 for 1, 3, 6, 10, 15 counters; the 6 for 6 counters is exactly the longest game you found today.” Martin Gardner wrote about this game in Scientific American in 1983 under the name Bulgarian solitaire, and that is the name to give the children at the end.

9. Reference data

All numbers below were computed by a program that lists every picture and applies the move; the theorems of Section 10 were checked against it for every $n \leq 24$.

The map for 6 counters

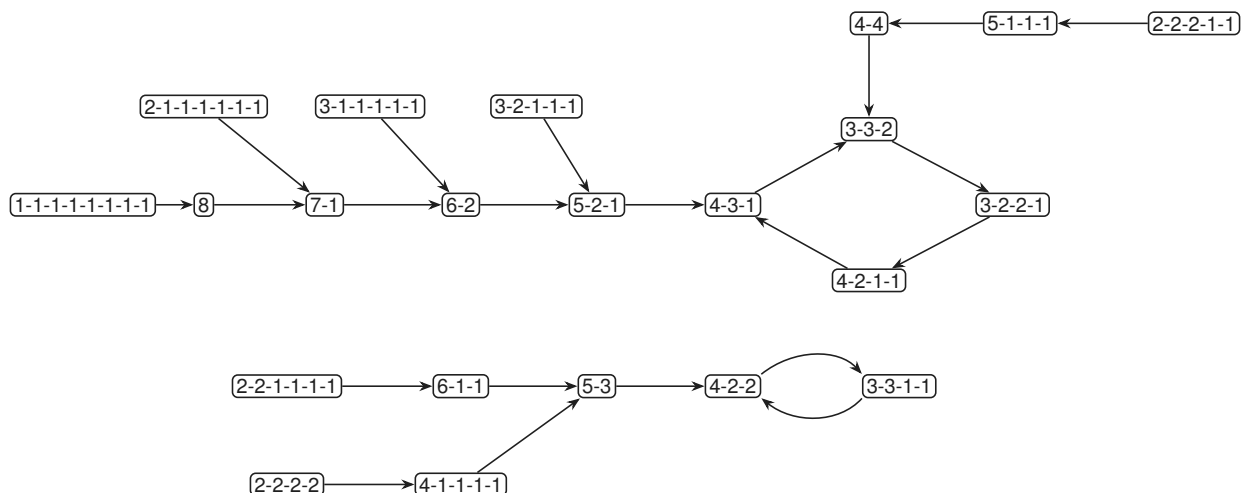


Sources (nothing leads to them): 2-2-1-1, 2-1-1-1-1, 1-1-1-1-1-1. Moves to the stairs: 6 from 2-2-1-1; 5 from 4-1-1; 4 from 3-3 and 1-1-1-1-1-1; 3 from 6, 2-2-2, 2-1-1-1-1; 2 from 5-1, 3-1-1-1; 1 from 4-2.

The maps for 3, 4, 5 and 7 counters

3 counters: $1-1-1 \rightarrow 3 \rightarrow 2-1$ (stuck). 4 counters: $1-1-1-1 \rightarrow 4 \rightarrow 3-1 \rightarrow 2-2 \rightarrow 2-1-1 \rightarrow 3-1$: one loop of length 3, no stuck picture. 5 counters: $1-1-1-1-1 \rightarrow 5 \rightarrow 4-1 \rightarrow 3-2 \rightarrow 2-2-1 \rightarrow 3-1-1 \rightarrow 3-2$, and $2-1-1-1 \rightarrow 4-1$: one loop of length 3. 7 counters, 15 pictures, one loop of length 4, $4-2-1 \rightarrow 3-3-1 \rightarrow 3-2-2 \rightarrow 3-2-1-1 \rightarrow$ back: feeders $1-1-1-1-1-1-1 \rightarrow 7 \rightarrow 6-1 \rightarrow 5-2 \rightarrow 4-2-1$; $2-1-1-1-1-1 \rightarrow 6-1$; $3-1-1-1-1 \rightarrow 5-2$; $2-2-1-1-1 \rightarrow 5-1-1 \rightarrow 4-3 \rightarrow 3-2-2$; $2-2-2-1 \rightarrow 4-1-1-1 \rightarrow 4-3$. Sources: 1-1-1-1-1-1-1, 2-1-1-1-1-1, 3-1-1-1-1, 2-2-1-1-1, 2-2-2-1.

The map for 8 counters



Loop A (length 4) collects 15 pictures, loop B (length 2) collects 7. Sources: 1-1-1-1-1-1-1-1, 2-1-1-1-1-1-1, 3-1-1-1-1-1, 3-2-1-1-1, 2-2-2-1-1, 2-2-1-1-1-1, 2-2-2-2. Longest run before a loop: 5 moves, from eight singles.

9 counters

30 pictures, one loop of length 4: $4-3-2 \rightarrow 3-3-2-1 \rightarrow 4-2-2-1 \rightarrow 4-3-1-1 \rightarrow$ back. Moves until the loop (next picture in parentheses). 1: 5-4 (4-3-2), 5-3-1 (4-3-2), 5-2-2 (4-3-1-1). 2: 6-3 (5-2-2), 6-2-1 (5-3-1), 6-1-1-1 (5-4), 5-1-1-1-1 (5-4), 4-2-1-1-1 (5-3-1), 3-3-1-1-1 (5-2-2). 3: 7-2 (6-2-1), 7-1-1 (6-3), 4-1-1-1-1-1 (6-3), 3-2-2-2 (4-2-1-1-1), 3-2-1-1-1-1 (6-2-1), 2-2-2-2-1 (5-1-1-1-1), 2-2-2-1-1-1 (6-1-1-1). 4: 8-1 (7-2), 3-3-3 (3-2-2-2), 3-1-1-1-1-1-1 (7-2),

2-2-1-1-1-1-1 (7-1-1). 5: 9 (8-1), 4-4-1 (3-3-3), 2-1-1-1-1-1-1 (8-1). 6: 5-2-1-1 (4-4-1), 1-1-1-1-1-1-1-1 (9). 7: 3-2-2-1-1 (5-2-1-1).

10 counters: all 42 pictures, by moves to the stairs

Next picture in parentheses; S marks a source (nothing leads to it).

0	4-3-2-1
1	5-3-2 (4-3-2-1)
2	6-4 (5-3-2); 6-3-1 (5-3-2); 4-3-1-1-1 (5-3-2)
3	7-2-1 (6-3-1); 7-1-1-1 (6-4); 5-1-1-1-1-1 (6-4); 4-2-2-2 (4-3-1-1-1); 4-2-1-1-1-1 (6-3-1) S
4	8-2 (7-2-1); 3-3-3-1 (4-2-2-2); 3-2-1-1-1-1-1 (7-2-1) S; 2-2-2-2-2 (5-1-1-1-1-1) S; 2-2-2-1-1-1-1 (7-1-1-1) S
5	9-1 (8-2); 4-4-2 (3-3-3-1); 3-1-1-1-1-1-1-1 (8-2) S
6	10 (9-1); 5-5 (4-4-2); 5-3-1-1 (4-4-2); 2-1-1-1-1-1-1-1 (9-1) S
7	6-2-2 (5-3-1-1); 6-1-1-1-1 (5-5); 4-2-2-1-1 (5-3-1-1); 1-1-1-1-1-1-1-1-1 (10) S
8	7-3 (6-2-2); 3-3-2-2 (4-2-2-1-1); 3-3-1-1-1-1 (6-2-2) S; 2-2-2-2-1-1 (6-1-1-1-1) S
9	8-1-1 (7-3); 4-3-3 (3-3-2-2); 4-1-1-1-1-1-1 (7-3) S
10	5-4-1 (4-3-3); 4-4-1-1 (4-3-3); 2-2-1-1-1-1-1-1 (8-1-1) S
11	6-2-1-1 (5-4-1); 5-2-2-1 (4-4-1-1); 5-2-1-1-1 (5-4-1)
12	3-3-2-1-1 (5-2-2-1) S; 3-2-2-2-1 (5-2-1-1-1) S; 3-2-2-1-1-1 (6-2-1-1) S

Number of starts by move count, 0 to 12: 1, 1, 3, 5, 5, 3, 4, 4, 4, 3, 3, 3, 3. Useful full games: 7-3 → 6-2-2 → 5-3-1-1 → 4-4-2 → 3-3-3-1 → 4-2-2-2 → 4-3-1-1-1 → 5-3-2 → 4-3-2-1 (8 moves); the three 12-move games join at 4-3-3 after three moves (3-2-2-2-1 → 5-2-1-1-1 → 5-4-1 → 4-3-3; 3-2-2-1-1-1 → 6-2-1-1 → 5-4-1 → 4-3-3).

Loops for 11 to 14 counters (stairs with 4 rows plus 1, 2, 3, 4 extras; all loops have length 5)

11: 5-3-2-1 → 4-4-2-1 → 4-3-3-1 → 4-3-2-2 → 4-3-2-1-1 → back. 12: 5-4-2-1 → 4-4-3-1 → 4-3-3-2 → 4-3-2-2-1 → 5-3-2-1-1 → back, and 5-3-3-1 → 4-4-2-2 → 4-3-3-1-1 → 5-3-2-2 → 4-4-2-1-1 → back. 13: 5-4-3-1 → 4-4-3-2 → 4-3-3-2-1 → 5-3-2-2-1 → 5-4-2-1-1 → back, and 5-4-2-2 → 4-4-3-1-1 → 5-3-3-2 → 4-4-2-2-1 → 5-3-3-1-1 → back. 14: 5-4-3-2 → 4-4-3-2-1 → 5-3-3-2-1 → 5-4-2-2-1 → 5-4-3-1-1 → back.

Summary table

n	pictures	k, r	loops (lengths)	loop pictures	longest run before the loop	sources
1	1	1, 0	1 (1)	1	0	0
2	2	1, 1	1 (2)	2	0	0
3	3	2, 0	1 (1)	1	2	1
4	5	2, 1	1 (3)	3	2	1
5	7	2, 2	1 (3)	3	3	2
6	11	3, 0	1 (1)	1	6	3
7	15	3, 1	1 (4)	4	4	5
8	22	3, 2	2 (4, 2)	6	5	7
9	30	3, 3	1 (4)	4	7	10
10	42	4, 0	1 (1)	1	12	14
11	56	4, 1	1 (5)	5	8	20
12	77	4, 2	2 (5, 5)	10	8	27
13	101	4, 3	2 (5, 5)	10	9	37
14	135	4, 4	1 (5)	5	14	49
15	176	5, 0	1 (1)	1	20	66
16	231	5, 1	1 (6)	6	15	86
17	297	5, 2	3 (6, 6, 3)	15	12	113
18	385	5, 3	4 (6, 6, 6, 2)	20	13	147
19	490	5, 4	3 (6, 6, 3)	15	16	190
20	627	5, 5	1 (6)	6	23	243
21	792	6, 0	1 (1)	1	30	311

$n = T_k + r$ with $0 \leq r \leq k$. Longest games to the stairs for T_k counters, $k = 1, \dots, 8$: 0, 2, 6, 12, 20, 30, 42, 56 $= k^2 - k$, attained by 1, 1, 1, 3, 16, 65, 293, 1267 starts (for $k = 5$, e.g. 4-4-3-2-1-1; for $k = 6$, e.g. 5-5-4-3-2-1-1). From the single pile: 0, 1, 3, 6, 10, 15, 21, 28 moves $= T_{k-1}$; from all singles one more (for $n \geq 2$).

10. The mathematics behind the session (for the two mathematicians)

The move is a map β on the set of partitions of n ; there are $p(n)$ of them, finite, so every orbit is eventually periodic (Q1: the pigeonhole argument the older children can give). The real questions are: which points are fixed (Q2), whether every orbit ends in the same place (Q3), how long that takes (Q4), which points have no preimage (Q5), and what the functional graph looks like (Q8).

Fixed points are exactly the staircases (Q2). Let B be the largest part of a fixed partition. After the move every old part is smaller than B , so the part of size B in the (identical) image is the new part; the new part has size equal to the number of parts, so $B = \text{number of parts}$. The old B became $B - 1$, which must therefore be present; it becomes $B - 2$, and so on down to 1: B parts of sizes $B, \dots, 1$. Conversely the staircase is fixed. So a fixed point exists iff n is triangular.

Brandt's theorem (Q3) and the diagonal potential. Draw a partition as a Young diagram, rows left-aligned, longest on top, and give the cell in row i , column j the diagonal number $i + j - 1$. Diagonal d has d cells. Let Φ be the sum of the diagonal numbers of all n cells. Perform the move in two steps. Step 1 ("unsorted move"): remove the first column and lay it as a new top row, the cell from row i going to column i ; every other cell slides down one row and left one column, $(i, j) \mapsto (i + 1, j - 1)$, and $(i, 1) \mapsto (1, i)$. Every cell keeps its diagonal number, so Φ is unchanged, and the rows are now $(m, \lambda_1 - 1, \lambda_2 - 1, \dots)$ with m the old number of parts. Step 2: sort the rows by swapping adjacent out-of-order rows. Swapping a row of length a sitting above a longer row of length b sends the a short-row cells down one diagonal and the b long-row cells up one: Φ changes by $a - b < 0$. Hence Φ never

increases, and stays fixed exactly when no sorting is needed, i.e. when $m \geq \lambda_1 - 1$. This is what Problem 8 of the grades 4–5 pages makes visible with cards whose rank is their diagonal number: through four unsorted moves every card rides its diagonal; at the fifth the sort costs one.

On a cycle Φ is constant, so every move on the cycle is already sorted after Step 1: every cell keeps its diagonal number forever and rotates along its diagonal, $(1, d) \rightarrow (2, d-1) \rightarrow \dots \rightarrow (d, 1) \rightarrow (1, d)$, with period d . Suppose diagonal $d+1$ carries a cell. Every $d+1$ moves it visits $(1, d+1)$, and since rows are left-aligned, $(1, d)$ is then occupied. The occupied set of diagonal d rotates with period d , and $\gcd(d, d+1) = 1$, so the moments $t \equiv t_0 \pmod{d+1}$ meet every rotation offset of diagonal d : diagonal d is full. By downward induction, a cycle partition consists of diagonals $1, \dots, k$ full plus $r \leq k$ cells on diagonal $k+1$, so $n = T_k + r$ with $0 \leq r \leq k$, and for triangular n the cycle is the staircase alone. This is Brandt's theorem (1982); the diagonal argument is the one in Akin–Davis, Étienne and Griggs–Ho.

Cycle partitions are “stairs plus wanderers”. Diagonals $1..k$ full plus r cells on diagonal $k+1$ means: the staircase $k, \dots, 1$ with a phantom empty row $k+1$, and r of these $k+1$ rows lengthened by one cell, at most one per row. Subtracting one from every row shifts the staircase down one row and carries each extra cell down one row; the new part has k cells plus one exactly when the phantom bottom row had an extra. So the extras walk down the stairs in lock-step, and the one that steps off the bottom reappears on top. The cycle length is the period of the extra-pattern under cyclic shift of $k+1$ rows, a divisor of $k+1$; the cycle partitions number $\binom{k+1}{r}$; the number of cycles is the number of binary necklaces of length $k+1$ with r ones (counting cycles is a first real use of Burnside). At $n = 8$: $k = 3$, $r = 2$, six cycle partitions, two necklaces, hence loops of lengths 4 (pattern rows 1,2) and 2 (pattern rows 1,3). At $n = 17$ ($k = 5$, $r = 2$) the pattern “rows 1,4” of six has period 3: loops of lengths 6, 6, 3.

How long (Q4). Igusa (1985) and Étienne (1991) proved that with T_k counters the staircase is reached within $k^2 - k$ moves, and the bound is sharp; the starts 2-2-1-1, 3-3-2-1-1, 4-4-3-2-1-1, ... (the staircase $k-1$ with an extra part $k-1$ and an extra part 1) attain it for every $k \leq 12$ tested. Two exact facts within the children's reach: from the single pile T_k the staircase arrives in exactly T_{k-1} moves, from n singles one move more (for $n \geq 2$). Proof of the first: after t moves the partition is a big part $T_k - t$ together with a small partition μ_t of t , where μ_{t+1} is the move applied to μ_t with the new part one bigger (the big part contributes a cell). By induction μ_{T_j+s} , $0 \leq s \leq j$, is the staircase j with its top s rows lengthened by one; at $t = T_{k-1}$, μ is the staircase $k-1$ and the big part is $T_k - T_{k-1} = k$: the staircase. Before that the big part exceeds k . The children see it in $10 \rightarrow 9-1 \rightarrow 8-2 \rightarrow 7-2-1 \rightarrow 6-3-1 \rightarrow 5-3-2 \rightarrow 4-3-2-1$.

Φ proves convergence, not speed: it is lazy. On the 6-move game from 2-2-1-1 it equals 15 for five moves and drops once, to 14. Values for 6 counters: 3-2-1: 14; 4-2, 4-1-1, 3-3, 3-1-1-1, 2-2-2, 2-2-1-1: 15; 5-1, 2-1-1-1-1: 17; 6, 1-1-1-1-1-1: 21.

Garden of Eden partitions (Q5). Undo a move: one part was new; it handed one cell back to every other part, and its leftovers were parts of size 1 that vanished. So it needs at least (number of parts) -1 cells. A partition has a preimage iff its largest part is at least (number of parts) -1 , i.e. iff its Dyson rank $\lambda_1 - \ell(\lambda)$ is at least -1 ; the number of preimages is the number of distinct part sizes $\geq \ell(\lambda) - 1$. Hopkins and Sellers (2007) enumerate the Garden of Eden partitions (rank ≤ -2): 1, 1, 2, 3, 5, 7, 10, 14 for $n = 3, \dots, 10$. The term comes from cellular automata (Moore's Garden of Eden theorem).

Counting (Q7) and structure (Q8). Pictures with n counters are the partitions $p(n)$: 1, 2, 3, 5, 7, 11, 15, 22, 30, 42. The functional graph is a forest of trees flowing into cycles; for $n = 6$ it is the Y of Section 9. For $n = 8$ the two basins (15 and 7 pictures) are not separated by the parity of the number of parts, of the largest part, by all-parts-even, or by the parity of Φ ; we know no simple rule, and say so.

Where it leads.

- J. Brandt, Cycles of partitions, *Proc. Amer. Math. Soc.* 85 (1982) 483–486: fixed point iff triangular, the cycle description, every start reaching the staircase. M. Gardner, Mathematical Games, *Scientific American*, August 1983, popularized the game under this name.
- The bound $k^2 - k$: K. Igusa, Solution of the Bulgarian solitaire conjecture, *Math. Mag.* 58 (1985) 259–271; G.

Étienne, Tableaux de Young et solitaire bulgare, *J. Combin. Theory Ser. A* 58 (1991) 181–197; also H. J. Bentz, Proof of the Bulgarian solitaire conjectures, *Ars Combin.* 23 (1987) 151–170. The readable exposition is E. Akin and M. Davis, Bulgarian solitaire, *Amer. Math. Monthly* 92 (1985) 237–250.

- Cycles as shifts and necklaces: J. R. Griggs and C.-C. Ho, The cycling of partitions and compositions under repeated shifts, *Adv. in Appl. Math.* 21 (1998) 205–227; B. Hopkins and M. A. Jones, Shift-induced dynamical systems on partitions and compositions, *Electron. J. Combin.* 13 (2006), R80.
- Garden of Eden partitions: B. Hopkins and J. A. Sellers, Exact enumeration of Garden of Eden partitions, *Integers* 7(2) (2007), A19.
- Survey: V. Drensky, The Bulgarian solitaire and the mathematics around it, arXiv:1503.00885 (2015): history, generalizations (Carolina and Austrian solitaire, σ -shifts of compositions), random Bulgarian solitaire (S. Popov, *Random Structures Algorithms* 27 (2005) 310–330) and the limit-shape results of Eriksson and Jonsson, with full references.
- Courses: integer partitions and Young's lattice (Andrews, *The Theory of Partitions*; Stanley, *Enumerative Combinatorics* I, §1.8); finite dynamical systems (functional graphs, transients, basins); monovariant arguments (Φ is a textbook one).
- For a strong student: characterize the basins at $n = 8$ or $n = 12$; explain why all three 12-move starts with 10 counters pass through 4-3-3; find the distribution of move counts to the staircase (for 10 counters: 1, 1, 3, 5, 5, 3, 4, 4, 4, 3, 3, 3, 3 starts take 0 to 12 moves) and its average; describe the 65 starts with 21 counters that take the full 30 moves.